

CSE 400: Fundamentals of Probability in Computing
Tutorial-2 Lecture Scribe

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Question 1

Problem Statement: A point is chosen at random on a line segment of length L . Interpret this statement and find the probability that the ratio of the shorter to the longer segment is less than $\frac{1}{4}$.

Step 1: Interpretation

Choosing a point at random on a line segment of length L means:

$$X \sim \text{Uniform}(0, L)$$

where X represents the distance of the chosen point from one endpoint. The segment is divided into two parts: Length X and Length $L - X$.

The shorter segment is:

$$\min(X, L - X)$$

The longer segment is:

$$\max(X, L - X)$$

We need:

$$\frac{\min(X, L - X)}{\max(X, L - X)} < \frac{1}{4}$$

Step 2: Case Analysis

Case 1: $X \leq L - X$

This occurs when:

$$X \leq \frac{L}{2}$$

Then:

$$\text{Shorter} = X, \quad \text{Longer} = L - X$$

Condition:

$$\frac{X}{L - X} < \frac{1}{4}$$

Multiply both sides by $(L - X > 0)$:

$$X < \frac{L - X}{4}$$

Multiply both sides by 4:

$$4X < L - X$$

$$5X < L$$

$$X < \frac{L}{5}$$

Case 2: $X > \frac{L}{2}$

Then:

$$\text{Shorter} = L - X, \quad \text{Longer} = X$$

Condition:

$$\frac{L - X}{X} < \frac{1}{4}$$

Multiply both sides by ($X > 0$):

$$L - X < \frac{X}{4}$$

Multiply both sides by 4:

$$4L - 4X < X$$

$$4L < 5X$$

$$X > \frac{4L}{5}$$

Step 3: Probability Computation

Favorable region:

$$X < \frac{L}{5} \quad \text{or} \quad X > \frac{4L}{5}$$

Total length of favorable region:

$$\frac{L}{5} + \frac{L}{5} = \frac{2L}{5}$$

Since total sample space length is L ,

$$P = \frac{2L/5}{L} = \frac{2}{5}$$

Question 2

(White region: $\mathcal{N}(4, 4)$, Black region: $\mathcal{N}(6, 9)$, Black fraction = α)

We are given reading $x = 5$.

Step 1: Densities at $x = 5$

White density:

$$\begin{aligned} f_W(5) &= \frac{1}{\sqrt{2\pi(4)}} \exp\left(-\frac{(5-4)^2}{2(4)}\right) \\ &= \frac{1}{2\sqrt{2\pi}} \exp\left(-\frac{1}{8}\right) \end{aligned}$$

Black density:

$$\begin{aligned} f_B(5) &= \frac{1}{\sqrt{2\pi(9)}} \exp\left(-\frac{(5-6)^2}{2(9)}\right) \\ &= \frac{1}{3\sqrt{2\pi}} \exp\left(-\frac{1}{18}\right) \end{aligned}$$

Step 2: Equal Error Condition

Probability of misclassifying white:

$$(1 - \alpha)f_W(5)$$

Probability of misclassifying black:

$$\alpha f_B(5)$$

Set equal:

$$(1 - \alpha)f_W(5) = \alpha f_B(5)$$

Substitute expressions:

$$(1 - \alpha)\frac{1}{2}e^{-1/8} = \alpha\frac{1}{3}e^{-1/18}$$

Solve for α :

$$\alpha = \frac{\frac{1}{2}e^{-1/8}}{\frac{1}{2}e^{-1/8} + \frac{1}{3}e^{-1/18}}$$

This is the required value of α .

Question 3

(a) $X \sim \text{Uniform}(0, A)$

We minimize:

$$E|X - a| = \int_0^A |x - a| \frac{1}{A} dx$$

Split at a :

$$= \frac{1}{A} \left[\int_0^a (a - x) dx + \int_a^A (x - a) dx \right]$$

Compute integrals:

$$\int_0^a (a - x) dx = \left(ax - \frac{x^2}{2} \right) \Big|_0^a = a^2 - \frac{a^2}{2} = \frac{a^2}{2}$$

$$\int_a^A (x - a) dx = \left(\frac{x^2}{2} - ax \right) \Big|_a^A = \left(\frac{A^2}{2} - aA \right) - \left(\frac{a^2}{2} - a^2 \right) = \frac{A^2}{2} - aA + \frac{a^2}{2}$$

Sum:

$$E|X - a| = \frac{1}{A} \left(a^2 - aA + \frac{A^2}{2} \right)$$

Differentiate with respect to a :

$$\frac{d}{da} = \frac{1}{A} (2a - A)$$

Set to zero:

$$\begin{aligned} 2a - A &= 0 \\ a &= \frac{A}{2} \end{aligned}$$

(b) $X \sim \text{Exponential}(\lambda)$

$$E|X - a| = \int_0^\infty |x - a| \lambda e^{-\lambda x} dx$$

Split at a . Differentiate w.r.t. a and set equal to zero. This yields the median condition:

$$P(X \leq a) = \frac{1}{2}$$

For exponential:

$$\begin{aligned} 1 - e^{-\lambda a} &= \frac{1}{2} \\ e^{-\lambda a} &= \frac{1}{2} \\ a &= \frac{\ln 2}{\lambda} \end{aligned}$$

Question 4

Mixture of exponentials.

We compute:

$$P(X > t + s \mid X > t) = \frac{P(X > t + s)}{P(X > t)}$$

Compute numerator:

$$P(X > t + s) = p_1 e^{-\lambda_1(t+s)} + p_2 e^{-\lambda_2(t+s)}$$

Denominator:

$$P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

Thus:

$$\frac{p_1 e^{-\lambda_1(t+s)} + p_2 e^{-\lambda_2(t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}$$

Question 5

Given $X \sim \text{Poisson}(\lambda_1)$, $Y \sim \text{Poisson}(\lambda_2)$.

We want:

$$P(X = k \mid X + Y = n)$$

Using definition:

$$= \frac{P(X = k, Y = n - k)}{P(X + Y = n)}$$

Numerator:

$$\frac{e^{-\lambda_1} \lambda_1^k}{k!} \cdot \frac{e^{-\lambda_2} \lambda_2^{n-k}}{(n-k)!}$$

Denominator:

$$\frac{e^{-(\lambda_1 + \lambda_2)} (\lambda_1 + \lambda_2)^n}{n!}$$

Simplify:

$$\binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{n-k}$$

Question 6

Transformation $Y = g(X)$, $X \sim \text{Uniform}[-2, 2]$.

CDF:

$$F_Y(y) = P(g(X) \leq y)$$

Find preimages and compute probabilities using:

$$P(a < X < b) = \frac{b-a}{4}$$

Differentiate:

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

(Graph construction follows from computed piecewise expressions.)

Question 7

Binomial model:

Number of up moves $S \sim \text{Bin}(1000, 0.52)$.

Need:

$$u^k d^{1000-k} \geq 1.3$$

Take log:

$$k \ln u + (1000 - k) \ln d \geq \ln 1.3$$

Solve for k .

Approximate using normal:

$$\mu = 1000(0.52) = 520$$

$$\sigma^2 = 1000(0.52)(0.48)$$

Standardize and compute probability.

Question 8

Given $T \sim \text{Exponential}(\lambda = 1/2)$.

(a)

$$P(T > 2) = e^{-(1/2)(2)} = e^{-1}$$

(b)

$$P(T > 10 \mid T > 9) = \frac{e^{-5}}{e^{-4.5}} = e^{-0.5}$$

Question 9

$$Z = \sqrt{X^2 + Y^2}$$

Use 2D CDF method:

$$F_Z(z) = \iint_{x^2+y^2 \leq z^2} f_X(x) f_Y(y) dx dy$$

Differentiate:

$$f_Z(z) = \int_0^{2\pi} f_X(z \cos \theta) f_Y(z \sin \theta) z d\theta$$

Question 10

Let $M_n = \max(X_1, \dots, X_n)$.

$$P(M_n \leq x) = F(x)^n$$

Joint probability:

$$P(M_n \leq x, M_{n+1} \leq y)$$

For $x \leq y$:

$$F(x)^n F(y)$$

For $x > y$:

$$F(y)^{n+1}$$

End of Complete Tutorial-2 Scribe